Example 5.16 Prove analytically that the line segment joining the mid-points of two sides of a triangle is parallel to the third side and is equal to half of its length.

Solution Let $P(a,b)$ $Q(c,d)$ and $R(e,f)$ be the vertices of a triangle.

Let S be the mid-point of PQ and T be the mid-point of PR

Therefore, $S = \left(\frac{a+c}{2}, \frac{b+d}{2}\right)$ and $T = \left(\frac{a+e}{2}, \frac{b+f}{2}\right)$

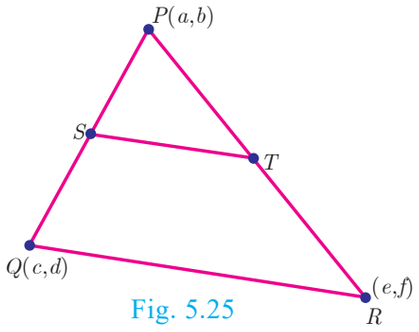


Fig. 5.25

Now, slope of $ST = \dfrac{\frac{b+f}{2} - \frac{b+d}{2}}{\frac{a+e}{2} - \frac{a+c}{2}} = \dfrac{f-d}{e-c}$

And slope of $QR = \dfrac{f-d}{e-c}$

Therefore, ST is parallel to QR. (since, their slopes are equal)

Also $ST = \sqrt{\left(\frac{a+e}{2} - \frac{a+c}{2}\right)^2 + \left(\frac{b+f}{2} - \frac{b+d}{2}\right)^2}$

$= \frac{1}{2}\sqrt{(e-c)^2 + (f-d)^2}$

$ST = \frac{1}{2}QR$

Thus ST is parallel to QR and half of it.

Note

This example illustrates how a geometrical result can be proved using coordinate Geometry.

Exercise 5.2

1. What is the slope of a line whose inclination with positive direction of x-axis is
   (i) $90°$ (ii) $0°$
2. What is the inclination of a line whose slope is (i) 0 (ii) 1
3. Find the slope of a line joining the points
   (i) $\left(5, \sqrt{5}\right)$ with the origin (ii) $(\sin\theta, -\cos\theta)$ and $(-\sin\theta, \cos\theta)$
4. What is the slope of a line perpendicular to the line joining $A(5,1)$ and P where P is the mid-point of the segment joining $(4,2)$ and $(-6,4)$.
5. Show that the given points are collinear: $(-3,-4)$, $(7,2)$ and $(12,5)$
6. If the three points $(3,-1)$, $(a,3)$ and $(1,-3)$ are collinear, find the value of a.
7. The line through the points $(-2,a)$ and $(9,3)$ has slope $-\frac{1}{2}$. Find the value of a.
8. The line through the points $(-2,6)$ and $(4,8)$ is perpendicular to the line through the points $(8,12)$ and $(x,24)$. Find the value of x.
9. Show that the given points form a right angled triangle.
   (i) $A(1,-4)$, $B(2,-3)$ and $C(4,-7)$ (ii) $L(0,5)$, $M(9,12)$ and $N(3,14)$

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